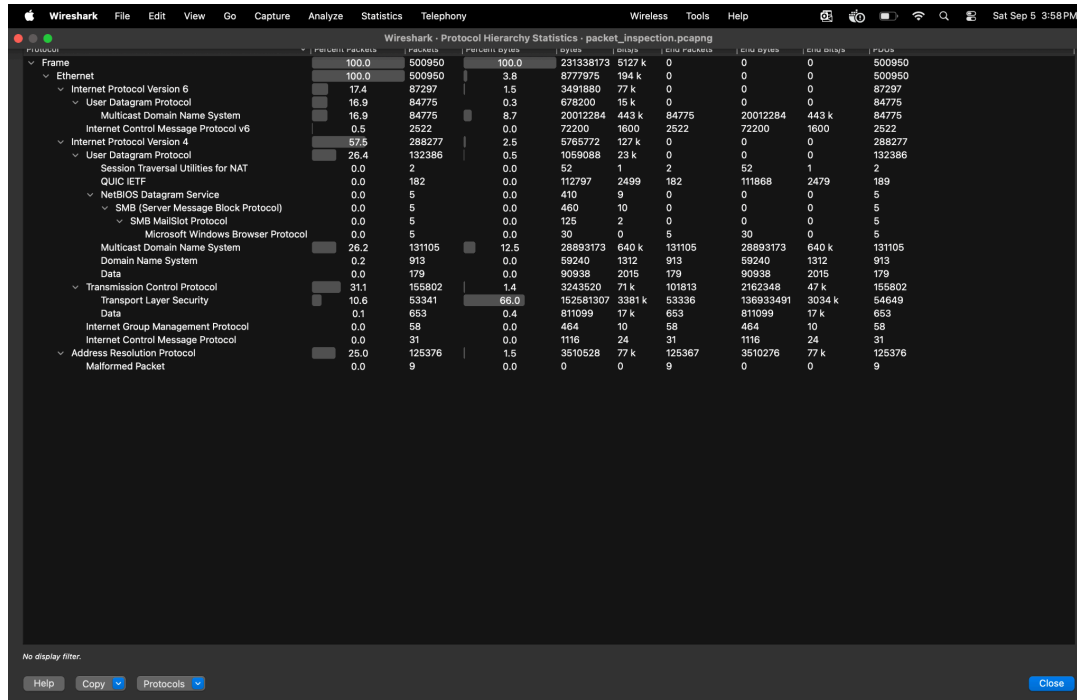


Ryan Fraser  
CTEC 445  
Professor Adrian Carter  
September 5, 2026

## Project 1 Wireshark



### 1. What protocols dominate traffic?

TCP dominated the traffic at 31.1%, followed by UDP at 26.4%. mDNS also accounted for 26.2% of the traffic. TLS dominated by data volume at 66.0% of total bytes.

| Endpoint Settings | Address | Packets | Bytes  | Tx Packets | Tx Bytes | Rx Packets | Rx Bytes | Country | City | Latitude | Longitude | AS Number | AS Organization |
|-------------------|---------|---------|--------|------------|----------|------------|----------|---------|------|----------|-----------|-----------|-----------------|
| 172.30.233.45     | 169,644 | 166 MB  | 38,663 | 8 MB       | 130,981  | 158 MB     |          |         |      |          |           |           |                 |
| 224.0.0.251       | 119,622 | 33 MB   | 0      | 0 bytes    | 119,622  | 33 MB      |          |         |      |          |           |           |                 |
| 172.217.152.241   | 50,742  | 59 MB   | 44,960 | 58 MB      | 5,792    | 484 kB     |          |         |      |          |           |           |                 |
| 142.251.155.4     | 7,853   | 6 MB    | 5,282  | 4 MB       | 2,561    | 2 MB       |          |         |      |          |           |           |                 |
| 13.36.76.35       | 35,077  | 43 MB   | 32,804 | 43 MB      | 2,163    | 143 kB     |          |         |      |          |           |           |                 |
| 3.167.81.139      | 8,349   | 9 MB    | 7,231  | 9 MB       | 1,118    | 109 kB     |          |         |      |          |           |           |                 |
| 142.251.151.119   | 4,029   | 4 MB    | 3,201  | 3 MB       | 828      | 416 kB     |          |         |      |          |           |           |                 |
| 172.253.139.119   | 2,589   | 2 MB    | 1,955  | 2 MB       | 634      | 52 kB      |          |         |      |          |           |           |                 |
| 23.39.184.112     | 4,348   | 5 MB    | 3,718  | 4 MB       | 630      | 263 kB     |          |         |      |          |           |           |                 |
| 10.0.255.51       | 923     | 98 kB   | 452    | 80 kB      | 471      | 38 kB      |          |         |      |          |           |           |                 |
| 192.178.155.94    | 1,359   | 1 MB    | 941    | 1 MB       | 418      | 41 kB      |          |         |      |          |           |           |                 |
| 162.247.243.29    | 698     | 444 kB  | 282    | 79 kB      | 416      | 365 kB     |          |         |      |          |           |           |                 |
| 172.217.152.130   | 1,369   | 1 MB    | 955    | 1 MB       | 414      | 45 kB      |          |         |      |          |           |           |                 |
| 74.125.155.161    | 2,558   | 3 MB    | 2,122  | 3 MB       | 406      | 39 kB      |          |         |      |          |           |           |                 |
| 172.217.155.138   | 2,779   | 3 MB    | 2,391  | 3 MB       | 388      | 38 kB      |          |         |      |          |           |           |                 |
| 151.101.66.24     | 2,475   | 3 MB    | 2,113  | 3 MB       | 362      | 188 kB     |          |         |      |          |           |           |                 |
| 142.251.154.119   | 1,081   | 835 kB  | 738    | 684 kB     | 343      | 151 kB     |          |         |      |          |           |           |                 |
| 173.184.53.189    | 1,376   | 1 MB    | 1,066  | 1 MB       | 310      | 53 kB      |          |         |      |          |           |           |                 |
| 142.250.31.132    | 825     | 565 kB  | 541    | 525 kB     | 284      | 40 kB      |          |         |      |          |           |           |                 |
| 23.39.184.104     | 686     | 428 kB  | 406    | 278 kB     | 280      | 150 kB     |          |         |      |          |           |           |                 |
| 44.215.132.129    | 421     | 250 kB  | 155    | 48 kB      | 266      | 202 kB     |          |         |      |          |           |           |                 |
| 172.217.152.210   | 1,543   | 2 MB    | 1,290  | 2 MB       | 253      | 29 kB      |          |         |      |          |           |           |                 |
| 172.64.155.209    | 480     | 292 kB  | 211    | 32 kB      | 249      | 260 kB     |          |         |      |          |           |           |                 |
| 216.239.34.157    | 681     | 463 kB  | 453    | 417 kB     | 228      | 47 kB      |          |         |      |          |           |           |                 |
| 64.233.180.132    | 470     | 241 kB  | 263    | 203 kB     | 207      | 38 kB      |          |         |      |          |           |           |                 |
| 74.125.155.135    | 2,068   | 2 MB    | 1,879  | 2 MB       | 189      | 34 kB      |          |         |      |          |           |           |                 |
| 74.125.155.225    | 884     | 901 kB  | 700    | 881 kB     | 184      | 20 kB      |          |         |      |          |           |           |                 |
| 3.171.01.160      | 840     | 798 kB  | 682    | 728 kB     | 178      | 70 kB      |          |         |      |          |           |           |                 |
| 63.140.36.38      | 326     | 154 kB  | 148    | 41 kB      | 178      | 113 kB     |          |         |      |          |           |           |                 |
| 172.253.115.132   | 868     | 883 kB  | 693    | 866 kB     | 175      | 17 kB      |          |         |      |          |           |           |                 |
| 192.179.18.94     | 281     | 62 kB   | 140    | 27 kB      | 141      | 36 kB      |          |         |      |          |           |           |                 |
| 173.184.53.41     | 1,690   | 2 MB    | 1,559  | 2 MB       | 131      | 14 kB      |          |         |      |          |           |           |                 |
| 192.158.11.132    | 291     | 64 kB   | 164    | 42 kB      | 127      | 42 kB      |          |         |      |          |           |           |                 |
| 172.217.152.245   | 1,293   | 1 MB    | 1,169  | 1 MB       | 124      | 25 kB      |          |         |      |          |           |           |                 |
| 142.250.31.139    | 257     | 101 kB  | 144    | 35 kB      | 113      | 67 kB      |          |         |      |          |           |           |                 |
| 172.30.200.174    | 349     | 78 kB   | 238    | 68 kB      | 111      | 10 kB      |          |         |      |          |           |           |                 |
| 172.30.194.18     | 398     | 109 kB  | 289    | 99 kB      | 109      | 10 kB      |          |         |      |          |           |           |                 |
| 173.184.7.70      | 841     | 919 kB  | 732    | 899 kB     | 109      | 20 kB      |          |         |      |          |           |           |                 |
| 172.30.200.72     | 368     | 91 kB   | 261    | 82 kB      | 107      | 9 kB       |          |         |      |          |           |           |                 |
| 142.251.111.102   | 345     | 277 kB  | 239    | 256 kB     | 106      | 22 kB      |          |         |      |          |           |           |                 |
| 172.217.135.167   | 950     | 1 MB    | 844    | 995 kB     | 106      | 27 kB      |          |         |      |          |           |           |                 |
| 172.30.240.61     | 480     | 150 kB  | 377    | 142 kB     | 103      | 9 kB       |          |         |      |          |           |           |                 |
| 172.30.234.106    | 416     | 89 kB   | 314    | 90 kB      | 102      | 9 kB       |          |         |      |          |           |           |                 |
| 172.30.243.59     | 415     | 135 kB  | 313    | 127 kB     | 102      | 9 kB       |          |         |      |          |           |           |                 |

## 2. What is the most frequent destination IP?

The most frequent destination IP address was 172.30.233.45, which received 130,981 packets during the capture.

| No.   | Time      | Source        | Destination | Protocol | Length | Info                                                     |
|-------|-----------|---------------|-------------|----------|--------|----------------------------------------------------------|
| 1457  | 1.626780  | 172.30.233.45 | 10.0.255.51 | DNS      | 81     | Standard query 0xc2c9 HTTP captive.g.aaplimg.com         |
| 1458  | 1.626941  | 172.30.233.45 | 10.0.255.51 | DNS      | 81     | Standard query 0xa350 A captive.g.aaplimg.com            |
| 1950  | 2.129156  | 172.30.233.45 | 10.0.255.51 | DNS      | 86     | Standard query 0xb07d A e0659.diced9.akamaiedge.net      |
| 3223  | 3.433837  | 172.30.233.45 | 10.0.255.51 | DNS      | 84     | Standard query 0xe5eb HTTPS s-0005.dual-s-msedge.net     |
| 3224  | 3.434004  | 172.30.233.45 | 10.0.255.51 | DNS      | 84     | Standard query 0xaa7 A s-0005.dual-s-msedge.net          |
| 17613 | 19.288390 | 172.30.233.45 | 10.0.255.51 | DNS      | 86     | Standard query 0x68bb HTTPS android.clients.google.com   |
| 17615 | 19.288462 | 172.30.233.45 | 10.0.255.51 | DNS      | 86     | Standard query 0xf8b A android.clients.google.com        |
| 18089 | 19.595788 | 172.30.233.45 | 10.0.255.51 | DNS      | 74     | Standard query 0xd150 HTTPS lh3.googleusercontent.com    |
| 18089 | 19.595858 | 172.30.233.45 | 10.0.255.51 | DNS      | 74     | Standard query 0x681a A lh3.googleusercontent.com        |
| 18028 | 19.633737 | 172.30.233.45 | 10.0.255.51 | DNS      | 88     | Standard query 0xe29a HTTPS ogads-pa.clients6.google.com |
| 18029 | 19.633801 | 172.30.233.45 | 10.0.255.51 | DNS      | 88     | Standard query 0xa189 A ogads-pa.clients6.google.com     |
| 18089 | 19.667163 | 172.30.233.45 | 10.0.255.51 | DNS      | 85     | Standard query 0xab9a HTTPS lh3.googleusercontent.com    |
| 18090 | 19.667214 | 172.30.233.45 | 10.0.255.51 | DNS      | 85     | Standard query 0x963e A lh3.googleusercontent.com        |
| 18308 | 19.859394 | 172.30.233.45 | 10.0.255.51 | DNS      | 86     | Standard query 0xd190 HTTPS encrypted-tbn0.gstatic.com   |
| 18311 | 19.859479 | 172.30.233.45 | 10.0.255.51 | DNS      | 86     | Standard query 0x6e2e A encrypted-tbn0.gstatic.com       |
| 18946 | 20.609837 | 172.30.233.45 | 10.0.255.51 | DNS      | 75     | Standard query 0x4acf HTTPS play.google.com              |
| 18947 | 20.609914 | 172.30.233.45 | 10.0.255.51 | DNS      | 75     | Standard query 0x6935 A play.google.com                  |
| 18951 | 20.611909 | 172.30.233.45 | 10.0.255.51 | DNS      | 78     | Standard query 0x2c84 HTTPS history.google.com           |
| 18952 | 20.611966 | 172.30.233.45 | 10.0.255.51 | DNS      | 78     | Standard query 0x7719 A history.google.com               |
| 19824 | 20.629512 | 172.30.233.45 | 10.0.255.51 | DNS      | 74     | Standard query 0xac76 HTTPS lh3.googleusercontent.com    |
| 19825 | 20.629567 | 172.30.233.45 | 10.0.255.51 | DNS      | 74     | Standard query 0x9839 A lh3.googleusercontent.com        |
| 19283 | 20.764382 | 172.30.233.45 | 10.0.255.51 | DNS      | 85     | Standard query 0x894e HTTPS lh3.googleusercontent.com    |

## 3. How many DNS queries were made?

There were 458 standard DNS query packets in the capture when mDNS and LLMNR traffic were excluded using the filter dns.flags.response == 0 and not mdns and not llmnr.